MATHS MANTHAN
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Subject: Mathematics **WEEKLY CHAPTER TEST – 01 (QUADRATIC EQUATIONS)** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. Which of the following is a quadratic equation? [1]
(A) $x^2 + 2x + 1 = (4 - x)^2 + 3$ (B) $-2x^2 = (5 - x)(2x - \frac{2}{5})$
(C) $(k + 1)x^2 + \frac{3}{2}x = 7$, where $k = -1$ (D) $x^3 - x^2 = (x - 1)^3$

Q2. Which of the following is NOT a quadratic equation? [1]
(A) $2(x - 1)^2 = 4x^2 - 2x + 1$ (B) $2x - x^2 = x^2 + 5$
(C) $(\sqrt{2}x + \sqrt{3})^2 + x^2 = 3x^2 - 5x$ (D) $(x^2 + 2x)^2 = x^4 + 3 + 4x^3$

Q3. Which of the following equations has 2 as a root? [1]
(A) $x^2 - 4x + 5 = 0$ (B) $x^2 + 3x - 12 = 0$ (C) $2x^2 - 7x + 6 = 0$ (D) $3x^2 - 6x - 2 = 0$

Q4. If $\frac{1}{2}$ is a root of the equation $x^2 + kx - \frac{5}{4} = 0$, then the value of k is: [1]
(A) 2 (B) -2 (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

Q5. **Assertion (A):** The equation $4x^2 - 12x + 9 = 0$ has two equal real roots.
Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has equal real roots if its discriminant $D = b^2 - 4ac = 0$. [1]
(A) Both A and R are true and R is correct explanation of A. (B) Both are true, but R is not correct explanation.
(C) A is true, R is false. (D) A is false, R is true.

SECTION B: Short Answer Type Questions (Part I)

Q6. Find the roots of the quadratic equation $6x^2 - x - 2 = 0$ by the method of factorization. [2]

Q7. The sum of the squares of two consecutive positive integers is 365. Find the integers. [3]

SECTION B: Short Answer Type Questions (Continued) (Total Sec B = 11 Marks)

Q8. Solve the following quadratic equation for x by factorization: [3]

$$4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$$

Q9. A natural number, when increased by 12, equals 160 times its reciprocal. Find the number. [3]

SECTION C: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q10. Case Study: Automotive Fuel Efficiency & Dual Vehicle Test Run

Two automobile test drivers, Johan and Jayant, conducted a highway performance trial over a track of 400 km. Johan drove Car A at a uniform cruising speed of x km/h. Jayant drove Car B at a speed that was 10 km/h faster than Johan's car. Because of the higher cruising speed, Jayant took 2 hours less than Johan to cover the entire 400 km distance.

Based on the above scenario, answer the following questions:

(i) Write expressions for the time taken by Johan and Jayant to cover 400 km in terms of x. [1]

(ii) Formulate the quadratic equation in x in standard form representing the difference in travel times. [1]

(iii) Find the speed of Johan's car (x) and the speed of Jayant's car. [2]

OR

Find the actual time taken by each car to complete the 400 km journey. [2]

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